1 Clarify

Confirmed request

Select ≤ 2 sites.
Maximize coverage.

Public: sites, demand
Fixed: budget, radius
Output: selected IDs

Clarifier
Facts + settings

2 Model

Typed binary formulation

x_i: site selected
y_j: demand covered

$$\max \Sigma_j w_j y_j$$

$$y_j \le \Sigma_i a_{ji} x_i$$

$$\Sigma_i x_i \le 2;\ x_i, y_j \in \{0,1\}$$

Optional pattern retrieval

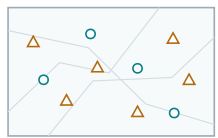
Modeler
Schema + builder

3 Prepare

Public sources → instance

site	lat	lon
s_1	…	…
s_2	…	…

Demand + radius → a_{ji}



Data Scout + Adapter

4 Solve

QUBO → binary candidate

$$E = f + \Sigma_k \lambda_k P_k$$

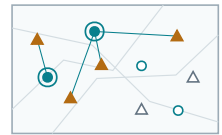


x = (1, 1, 0, 0)

Compiler + backend
Preflight + bounded calls

5 Deliver

Original-constraint checks



● Selected site
▲ Covered demand

Visualizer + explanation
Decision + provenance

Request + data schema

Requirement Checker
Separate feasibility / ranking evaluator

Data Checker
Independent counts / totals

Delivery status
Usable / limited / unavailable

Validation evidence → bounded data or model repair

Schematic coverage example; checkers provide evidence, not a correctness guarantee.

A Data interpretation

ID	value	unit
v_1	120	people
v_2	80	people

Raw-table profiles
Request + schema

Adapter
Code on full source tables

Data Checker
Separate count / total code
Checker does not see Adapter code

≈

Counts + numeric totals
0.5% tolerance; unit-scale floor
Reducible-set counts skipped

B Requirement consistency

Confirmed request
≤ 5 sites; count once
Schema + decision IDs

Modeler
Formulation + decision keys

Requirement Checker
Evaluator + decision sampler
Checker does not see formulation

≈

Feasibility + ranking
Fix decisions; optimize helpers
Timeout / unsupported: inconclusive
Also evaluate delivered decisions

C External acceptance — evaluation only

Frozen raw-data truth
Withheld from runtime agents

→

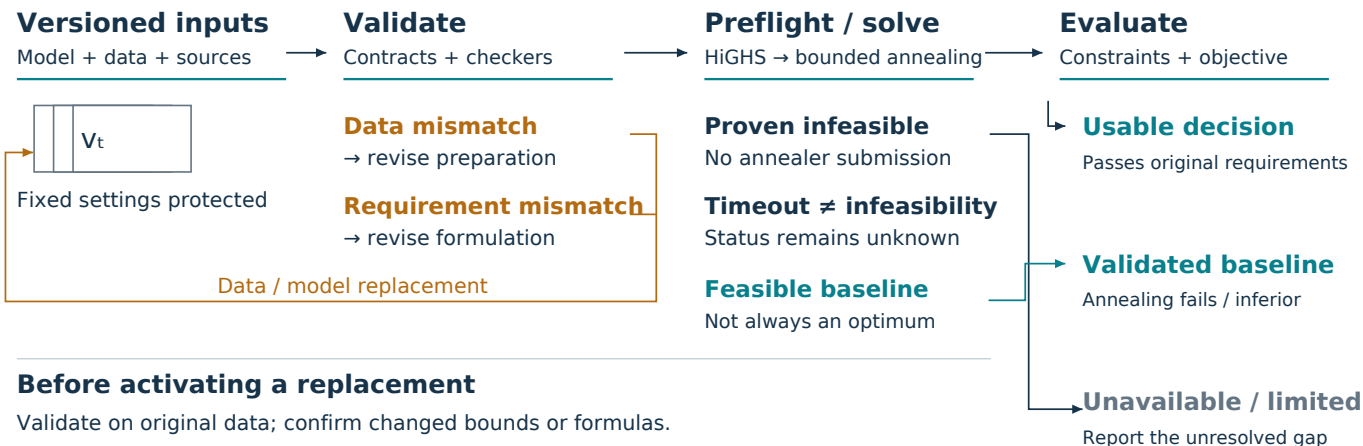
Case-specific judge
Prepared data, model, delivery

→

① Data ② Model ③ Delivery

Joint success requires all three gates

Runtime agreement is evidence; shared LLM/schema errors can remain. The judge is not a repair oracle.



Before activating a replacement

Validate on original data; confirm changed bounds or formulas.
Retain earlier usable version; tie every solution to its model/data version.
Finite repair attempts, solver calls and process timeouts.

Solver failure alone does not establish a model error; penalty tuning remains a bounded inner loop.

A Development cases

Taipei chargers
Maximum coverage; two scales

YouBike
Closed tour; two scales

Taipei AED
Balanced partition; two scales

London ambulance
p-median; two scales

Berlin playgrounds
Incremental coverage; two scales

Fresh runs; no saved model or cache

B Mechanism ladder

1. **Direct LLM**
   Whole-system baseline
2. **Structured workflow**
   Typed interfaces + toolkit
3. **Workflow with checkers**
   Add separate computations
4. **Complete workflow**
   Add cross-stage repair
5. **Workflow with retrieval**
   Add modelling patterns

Without clarification web lookup
Acquisition still searches the web

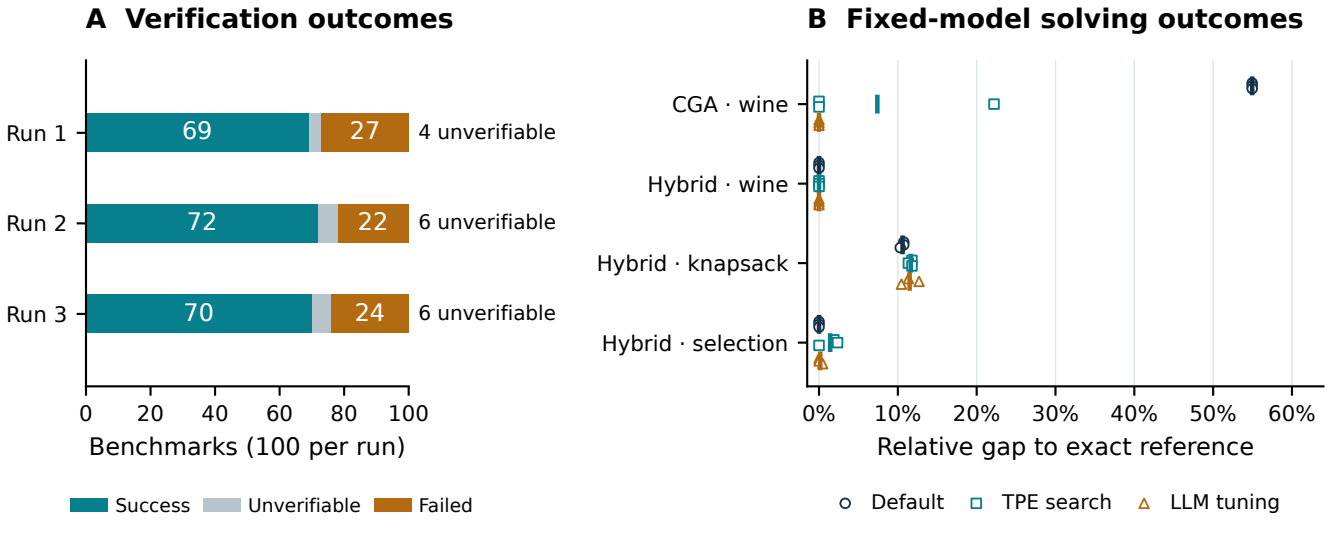
C External judging

Truth withheld from agents

1. **Data correct**
   Entities, coordinates, values
2. **Model correct**
   Feasibility + objective ranking
3. **Usable delivery**
   Requirements + reference quality

Joint success = all three
Three repetitions planned

Protocol, not completed results. Freeze code, prompts, cases, judge and truth; direct tools differ.


A Verification outcomes
Run 1
Run 2
Run 3
69
27
4 unverifiable
72
22
6 unverifiable
70
24
6 unverifiable
0
20
40
60
80
100
Benchmarks (100 per run)
Success
Unverifiable
Failed
B Fixed-model solving outcomes
CGA · wine
Hybrid · wine
Hybrid · knapsack
Hybrid · selection
0%
10%
20%
30%
40%
50%
60%
Relative gap to exact reference
Default
TPE search
LLM tuning